

# Dr Michèle Christopher Renard

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## RESEARCH PROFILE

Environmental physiologist and quantitative researcher working at the intersection of heat exposure, sleep and population health. Currently Research Fellow at the National University of Singapore, where I provide analytical leadership across two national research programmes — Project HEATS and Cooling Singapore 2.0 — modelling how the indoor thermal environment shapes sleep and physiological strain in free-living tropical populations. My work spans large longitudinal wearable and environmental datasets, randomised cooling interventions, and evidence synthesis that feeds directly into heat-health advisory policy. I hold a PhD in Sport, Health and Exercise Sciences, publish across environmental physiology and nutritional science, supervise three doctoral candidates and a postdoctoral researcher, and lead strategic industry partnerships for the NUS Medicine Early Career Researchers Office.

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## PROFESSIONAL EXPERIENCE

**Research Fellow** — National University of Singapore, Yong Loo Lin School of Medicine

*Human Potential Translational Research Programme (HPTRP) & Heat Resilience and Performance Centre (HRPC), Singapore | Apr 2025 – Present*

- Provide analytical leadership across two NRF-funded programmes — **Project HEATS** (Heat Exposure, Activity and Sleep) and **Cooling Singapore 2.0 (CS2.0)** — delivering Python-based statistical modelling for longitudinal, observational and intervention analyses.
- Lead the primary analysis of bedroom thermal exposure and sleep continuity across **122 participants and 10,303 person-nights** collected over three field waves (Jun 2025 – Jul 2026), applying linear mixed-effects models with Mundlak within/between decomposition to separate behavioural adaptation from physiological strain.
- Develop and apply non-linear threshold methods — piecewise/segmented models with AIC-based knot search, and restricted cubic splines — to compare dry-bulb temperature against composite indices (wet-bulb temperature, Heat Index) as predictors of sleep disruption.
- Design analysis plans and deliver statistical analysis and scientific reporting for (i) a longitudinal study of indoor heat exposure and sleep outcomes and (ii) a randomised intervention evaluating low-energy cooling strategies in migrant workers' dormitories.
- **CS2.0**: model laboratory-derived physiological responses to controlled heat exposure across population groups (children, heat-unacclimatised adults, older adults) and translate modelled outputs into policy-relevant insight for the tailoring of national heat-health advisories.
- Lead and co-lead four systematic reviews spanning heat-health advisory effectiveness, heat vulnerability index development, warm-night temperature thresholds, and fluid temperature/form in exercise cooling.
- Supervise and mentor doctoral and postdoctoral researchers in analysis planning, coding practice, interpretation and scientific writing; coordinate across a cross-institutional team spanning NUS, UC Berkeley and BEARS.
- Research from Project HEATS attracted national media coverage in 2026 in *The Straits Times* and *Lianhe Zaobao*.

**Postdoctoral Research Associate** — King's College London, London, UK

*Department of Nutritional Sciences | Jul 2024 – Apr 2025*

- Led and coordinated research on cooking and food behaviours during the cost-of-living crisis, managing participant recruitment, data collection and study documentation.
- Conducted qualitative and quantitative analysis, ensuring accurate data management and timely dissemination through manuscript preparation.
- Collaborated with multidisciplinary teams to develop research protocols and conduct field-based studies, surveys and scoping reviews.
- Presented findings at project meetings and maintained ongoing stakeholder collaboration.

**Research Assistant** — King's College London, London, UK

*Department of Nutritional Sciences | Feb 2024 – Jul 2024*

- Registered and led a scoping review on food, nutrition and sustainability-focused living laboratories in higher education and their impact on student learning.
- Conducted statistical analysis of large survey datasets in SPSS (chi-square, Kruskal–Wallis, hierarchical regression) examining food insecurity trends and evaluating culinary nutrition education interventions.

- Performed thematic analysis of focus group transcripts and open-text survey responses, working as a mixed-methods researcher across both workstreams.
- Designed and delivered a survey evaluating student and faculty readiness for a more coding-intensive statistics curriculum in Nutrition & Dietetics.

**Assistant Lecturer in Exercise Physiology** — Atlantic Technological University, Sligo, Ireland

*Jan 2022 – Jun 2022*

- Lecturer for Introduction to Kinesiology and Exercise Instruction, Human Anatomy & Physiology, Exercise Physiology, and Work Placement Preparation.
- Research project supervisor for undergraduate students across BSc Health Science and Physical Activity and BSc Health Science and Physiology.

**Assistant Lecturer in Sport & Exercise Nutrition** — Atlantic Technological University, Sligo, Ireland

*Oct 2021 – Jan 2022*

- Module lead and lecturer for the MSc Sports and Exercise Nutrition advanced performance nutrition module, covering diverse sports, special populations and nutrition in extreme environments.
- Lecturer for undergraduate modules in human nutrition and functional anatomy & kinesiology; provided academic input on course development and participated in programme committees.

**Postgraduate Researcher** — Technological University of the Shannon, Athlone, Ireland

*Oct 2019 – May 2024*

- Investigated determinants of dietary intake among team sport athletes to inform behavioural interventions, including tailored nutrition education and the use of training centres as venues for teaching culinary and nutritional management skills.
- Supported teaching and delivery across undergraduate modules in research methods, human physiology, fundamentals of sports science, sports nutrition, and strength and conditioning; graded assessments and provided formative feedback.
- Delivered presentations and workshops on nutrition and sports performance to sporting organisations, coaches and athletes.
- Provided sport science services at the University's sports arena — coaching, strength and conditioning, and performance testing (VO<sub>2</sub>max, lactate threshold, speed, 1RM).

**Exercise Instructor** — Technological University of the Shannon, Athlone, Ireland

*Aug 2019 – Mar 2024 (part-time)*

- Delivered gym inductions, fitness assessments and individualised programming; instructed group exercise across a range of abilities.
- Performed performance testing for teams and individual athletes; delivered athletics coaching.

**Health Advisor** — Bupa, United Kingdom

*Mar 2018 – Jul 2019*

- Delivered tailored clinical health assessments including anthropometry, blood pressure, **12-lead ECG**, point-of-care blood analysis for HbA<sub>1c</sub>, haemoglobin and lipid profile (**trained phlebotomist**), **spirometry** (flow–volume loop), functional movement screening and **sub-maximal cardiopulmonary exercise testing**.
- Provided individual lifestyle coaching using motivational interviewing, working autonomously and within a multi-disciplinary team of health advisors, physiotherapists, doctors and clinical administrators.

**Personal Trainer** — Nuffield Health, United Kingdom

*Aug 2017 – Mar 2018*

- Delivered one-to-one personal training, programming and progress review; instructed group exercise classes.
- Performed community-based health assessments (anthropometry, blood pressure, fasting blood glucose) to identify chronic disease risk and define achievable lifestyle goals.

**Research Assistant (Summer Internship)** — Kingston University, London, UK

*May 2016 – Sep 2016*

- Paid internship on a **British Academy / NHS-funded** project investigating exercise as a prescribed auxiliary treatment for patients living with HIV.
- Conducted statistical analysis of health, fitness and quality-of-life outcomes, and managed data and reporting for a large-scale longitudinal physical activity intervention.

## RESEARCH FUNDING

**Climate Impact Science Research (CISR) Programme** — National Research Foundation, Singapore & National Environment Agency

*Award No. CISR-2024-2R-04 — Exertional Heat Vulnerability Index (eHVI) for Singapore*

Research team member leading the evidence-synthesis workstream. Led development of the systematic review protocol underpinning the index — search strategy, data extraction framework and synthesis approach — and serve as **guarantor of the review**. Chair the bi-weekly technical review and act as quality gate for report blocks delivered to the funder.

**Campus for Research Excellence and Technological Enterprise (CREATE)** — National Research Foundation, Prime Minister's Office, Singapore

*Project HEATS — Heat Exposure, Activity and Sleep*

Named contributor delivering the statistical analysis plan and leading manuscript development across the heat exposure workstream.

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## RESEARCH SUPERVISION & MENTORSHIP

**Raagavi Mani** — PhD candidate, National University of Singapore

*Nighttime heat and humidity and sleep among outdoor workers in non-air-conditioned, multi-occupancy dormitories*. Named supervisor in the manuscript author contributions; co-author on the analysis and on two ICEE 2026 abstracts.

**Gabriel May Moe Kyaw** — PhD candidate, National University of Singapore

*A Systematic Review of Adverse Nighttime Heat Thresholds and Associated Effects on Sleep, Comfort and Health* (177 records). PROSPERO CRD420251009764. Second author; reviewed drafts V1–V3.

**Jessica Loke (Mei Qi)** — PhD candidate, National University of Singapore

*The Role of Fluid Temperature and Form in Cooling: A Systematic Review and Meta-Analysis* (60 studies, 644 participants, 41 in meta-analysis). PROSPERO CRD420251134107. Second author; delivered two full review rounds and adjudicated the Risk of Bias 2.0 assessments.

**Dr Warinthorn Angkanasirikul** — postdoctoral researcher, National University of Singapore

eHVI technical report (CISR-2024-2R-04). Chair the bi-weekly review cycle and provide structured methodological review of each report block prior to funder submission.

**Undergraduate research supervision** — Atlantic Technological University, 2021–2022

Supervised ten undergraduate research projects on topics including hamstring-to-quadriceps ratios in team athletes, dietary intake in female football players, and mental health in athletes.

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## TEACHING & INVITED WORKSHOPS

**“Which Test, and Why?” — CLIF Foundational Workshop: Statistical Approaches**

National University of Singapore | **28 August 2026** (60 min) — *forthcoming*

Invited presenter for one of four 2026 CLIF Foundational Workshops. A decision-first primer on statistical test selection for junior research assistants and coordinators. Accompanied by a published open-access interactive test chooser ([michelerenard.github.io/clif-chooser](https://michelerenard.github.io/clif-chooser)) and a one-page reference sheet, with worked examples in both R and Python.

**Project HEATS Interactive Workshop** — Indoor Air 2026, Singapore | 15 June 2026

Featured speaker (with S. Schiavon, R. Mani, W. Zhang and X. Fan) presenting the HEATS framework for optimising sleep — heat exposure characterisation, protection of vulnerable groups, technical cooling and behavioural intervention.

**MSc Sports and Exercise Nutrition — Advanced Performance Nutrition** (module lead), Atlantic Technological University, 2021–2022.

**Undergraduate teaching**, Atlantic Technological University and Technological University of the Shannon, 2019–2022 — Human Nutrition, Exercise Physiology, Human Anatomy & Physiology, Kinesiology, Exercise Instruction, Research Methods, Strength & Conditioning. Delivered and assessed laboratory sessions in functional movement screening, metabolic assessment (resting metabolic rate, energy expenditure) and performance testing (VO<sub>2</sub>max, lactate threshold).

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## PEER REVIEW & PROFESSIONAL SERVICE

### Invited Journal Reviewer

| Journal                                                           | Year |
|-------------------------------------------------------------------|------|
| <i>Sports Medicine – Open</i>                                     | 2026 |
| <i>Environmental Science and Pollution Research International</i> | 2025 |
| <i>Journal of Science in Sport and Exercise</i>                   | 2021 |
| <i>Journal of the International Society of Sports Nutrition</i>   | 2021 |

## LEADERSHIP & VOLUNTARY SERVICE

### Lead — Strategic Partnerships, Business Development & External Relations

NUS Medicine Early Career Researchers Office (ECRO) | 1 September 2026 – 31 August 2027

- Hold one of five ECRO Lead portfolios, with primary responsibility for business development and industry engagement on behalf of the School's early career researcher community.
- Represent NUS Medicine ECRO in partnership negotiations with a global pharmaceutical clinical research partner, resulting in an agreed pilot placement programme for postgraduate researchers from Q1 2027.
- Contributed to the review of portfolio scope, resourcing and governance across the Leads group, proposing a shared partner register to coordinate industry-facing activity across overlapping portfolios.

**Education Subcommittee Lead** — NUS Yong Loo Lin School of Medicine Postdoctoral Association (PDA) | Sep 2025 – Present

- Support the planning and delivery of the annual PDA Day.
- Coordinate panellists, speakers and external facilitators for professional development sessions; moderate panel discussions and facilitate session flow.
- Manage procurement, budgeting and related financial administration for events.
- Assist with the design, promotion and dissemination of professional development activities and resources for early career researchers.

**Research Liaison Officer** — Sport and Exercise Nutrition Group, Irish Nutrition and Dietetic Institute | Feb 2021 – Sep 2022

- Contributed to a monthly research newsletter for all members and shared CPD opportunities.
- Promoted collaborative research between academics and local practitioners; provided academic insight into resources shared with members.

## EDUCATION

**PhD — Sport, Health and Exercise Sciences** | Technological University of the Shannon, Athlone, Ireland | Oct 2019 – Jun 2024

Thesis: *Determinants of Dietary Intake in Team Sport Athletes.*

- Assessed athletes' dietary intake against sports nutrition guidelines and provided feedback for improvement.
- Evaluated athletes' nutrition knowledge and cooking skills to identify focus areas for educational workshops.
- Conducted interviews with athletes to explore barriers and facilitators to meeting dietary needs, informing holistic behaviour change interventions.

**MSc — Applied Sports Nutrition** | St Mary's University Twickenham, London, UK | Sep 2017 – Sep 2019

Thesis: *Evaluating Nutrition Knowledge in Elite and Sub-Elite Gaelic Football Players.*

Key modules: Advanced Sports Nutrition; Anthropometry and Body Composition Management; Research Methods; Statistics.

**BSc (Hons) — Exercise, Nutrition and Health** | Kingston University, London, UK | Sep 2014 – Jun 2017

Thesis: *Does a 4-Week Eucaloric Low-Carb, High-Fat Diet Affect the Strength Performance of Active, Normal-Weight Males?*

Key modules: Sport and Exercise Physiology; Extreme Environments and Ergogenic Aids; Sport and Exercise Psychology.

## HONOURS & AWARDS

**JSPS HOPE Fellow** | Japan Society for the Promotion of Science | Mar 2026

- Awarded on completion of the 17th JSPS HOPE Meeting with Nobel Laureates, Tsukuba, Japan.
- Recognised for excellence in both individual poster and multidisciplinary team presentations.
- Selected to represent the NUS Yong Loo Lin School of Medicine, building international collaborative networks with Nobel Laureates and early career researchers across the Asia-Pacific and Africa regions.

**Winner, Postgraduate Competition** | The Nutrition Society (UK & Ireland) | Jul 2024

- Competitive award providing a 20-minute invited lecture at the International Congress in Belfast.
  - Delivered “*Optimising dietary behaviour: can sports nutrition research contribute to improvements in population health?*”, critically evaluating the translation of sports nutrition research into public health strategy.
  - Applied hierarchical regression modelling to identify nutrition knowledge and culinary skills as key modifiable determinants of behaviour, and advocated for the integration of big data and machine learning in systems-based nutritional policy.
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## PUBLICATIONS

*Eleven peer-reviewed publications; three manuscripts under review; six in preparation.*

### Peer-Reviewed Publications

1. George, C. V., Bell, Z., **Renard, M.**, Brown, E., Kloskowska, P., Edwards, L., Newman, A., Rauber, A., Flynn, A. C., & Lavelle, F. (2026). Pregnancy and postpartum nutritional wants and needs in athletes: A cross-sectional survey in the United Kingdom and Republic of Ireland. *Performance Nutrition*, **2**(1), 14. <https://doi.org/10.1186/s44410-026-00030-0>
2. **Renard, M.**, Bell, Z., Jamshidvand, M., Mai, Z., McCloat, A., Mooney, E., Hollywood, L., & Lavelle, F. (2025). Domestic cooking and food behaviours during the COVID-19 pandemic and the cost-of-living crisis: A scoping review. *Appetite*, **216**, 108311. <https://doi.org/10.1016/j.appet.2025.108311>
3. Camp, R., Witard, O. C., **Renard, M.**, Ó Catháin, C., & Lavelle, F. (2025). Do athletes cook? A systematic scoping review of culinary nutrition in athletes. *Nutrition Bulletin*. <https://doi.org/10.1111/nbu.70012>
4. Buckley, J., Araujo, J. A., ... **Renard, M.**, ... Zainol, I. (2024). How transparent are quantitative studies in contemporary technology education research? Instrument development and analysis. *International Journal of Technology and Design Education*, **34**(2), 461–483. <https://doi.org/10.1007/s10798-023-09827-9>
5. **Renard, M.**, Kelly, D. T., Ní Chéilleachair, N., Lavelle, F., & Ó Catháin, C. (2023). Cooking and food skills confidence of team sport athletes in Ireland. *Nutrition Bulletin*, **48**(3), 329–342. <https://doi.org/10.1111/nbu.12625>
6. Taim, B. C., Ó Catháin, C., **Renard, M.**, Elliott-Sale, K. J., Madigan, S., & Ní Chéilleachair, N. (2023). The prevalence of menstrual cycle disorders and menstrual cycle-related symptoms in female athletes: A systematic literature review. *Sports Medicine*, **53**(10), 1963–1984. <https://doi.org/10.1007/s40279-023-01871-8>
7. Buckley, J., Adams, L., ... **Renard, M.**, ... Zhang, L. (2022). An assessment of the transparency of contemporary technology education research employing interview-based methodologies. *International Journal of Technology and Design Education*, **32**(4), 1963–1982. <https://doi.org/10.1007/s10798-021-09695-1>
8. **Renard, M.**, Anton-Solanas, A., Kelly, D. T., & Ó Catháin, C. (2022). Evaluation of nutrition knowledge in elite and sub-elite Gaelic football players. *Science and Medicine in Football*, **6**(1), 82–88. <https://doi.org/10.1080/24733938.2021.1883195>
9. **Renard, M.**, Kelly, D. T., Ní Chéilleachair, N., & Ó Catháin, C. (2021). How does the dietary intake of female field-based team sport athletes compare to dietary recommendations for health and performance? A systematic literature review. *Nutrients*, **13**(4), 1235. <https://doi.org/10.3390/nu13041235>
10. **Renard, M.**, Kelly, D. T., Ní Chéilleachair, N., & Ó Catháin, C. (2020). Evaluation of nutrition knowledge in female Gaelic games players. *Sports*, **8**(12), 154. <https://doi.org/10.3390/sports8120154>
11. Ó Catháin, C., Fleming, J., **Renard, M.**, & Kelly, D. T. (2020). Dietary intake of Gaelic football players during game preparation and recovery. *Sports*, **8**(5), 62. <https://doi.org/10.3390/sports8050062>

## Manuscripts Under Review

- **Renard, M.**, Frei, M., Chua, Y. X., Zhang, W., Parkinson, T., Lo, J. C.-Y., Lee, J. K. W., Schiavon, S., & Miller, C. HEATS Exposure: Protocol for wearable and in-home monitoring of indoor environment, sleep and activity in Singapore. *BMC Public Health*. Submitted May 2026; under review.
- **Renard, M.**, Lavelle, F., Kelly, D. T., Ní Chéilleachair, N., & Ó Catháin, C. Dietary behaviours in Gaelic football: A mixed-methods exploration of players' capability, opportunity and motivation. *Performance Nutrition* (special issue: *Optimizing Performance and Health through Sport Nutrition — A Behavioural Approach*). Revised and resubmitted July 2026; second round of review.
- **Renard, M.**, et al. Food insecurity, food behaviours, diet quality and mental wellbeing in the UK during the cost-of-living crisis. *Public Health Nutrition*. Under review.

## Manuscripts in Preparation

- **Renard, M.**, et al. Nighttime bedroom temperature and sleep continuity in Singapore homes. *Primary analysis of 122 participants across 10,303 person-nights*.
- **Renard, M.**, Chua, Y. X., Tan, P. M. S., Ioannou, L., Kingma, B., Morabito, M., Gao, C., & Lee, J. K. W. A systematic review of heat health advisory tools and their effectiveness (113 studies). PROSPERO **CRD420251052384**.
- **Renard, M.**, Angkanasirikul, W., Tan, P. M. S., Law, L. Y. L., & Lee, J. K. W. A systematic review to inform heat vulnerability index development in urban settings. PROSPERO **CRD420261289729**.
- Kyaw, G. M. M., **Renard, M.**, Fan, X., & Lee, J. K. W. A systematic review of adverse nighttime heat thresholds and associated effects on sleep, comfort and health. PROSPERO **CRD420251009764**.
- Loke, J. M. Q., **Renard, M.**, Alhadad, S. B., & Lee, J. K. W. The role of fluid temperature and form in cooling: A systematic review and meta-analysis. PROSPERO **CRD420251134107**. Target: *Sports Medicine*.
- Mani, R., Tan, S. C. C., **Renard, M.**, Frei, M., Loke, J. M. Q., Leow, C. H. W., Tan, P. M. S., Parkinson, T., Lo, J. C.-Y., Schiavon, S., & Lee, J. K. W. Nighttime heat and humidity are associated with impaired sleep among outdoor workers in non-air-conditioned, multi-occupancy dormitories.
- **Renard, M.**, et al. Physiological responses to controlled heat exposure in unacclimatised adults (Cooling Singapore 2.0).

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## REGISTERED REVIEW PROTOCOLS (PROSPERO)

| Registration           | Review                                                                                   | Role                                         |
|------------------------|------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>CRD420261289729</b> | Heat vulnerability index development in urban settings                                   | Lead author; protocol lead; <b>guarantor</b> |
| <b>CRD420251052384</b> | Heat health advisory tools and effectiveness (113 studies)                               | Lead author                                  |
| <b>CRD420251009764</b> | Adverse nighttime heat thresholds and effects on sleep, comfort and health (177 records) | Second author; supervisor                    |
| <b>CRD420251134107</b> | Fluid temperature and form in cooling (60 studies, 644 participants)                     | Second author; supervisor                    |

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## CONFERENCE PRESENTATIONS

### Delivered

**Indoor Air 2026** — Singapore, 14–18 June 2026 (*full paper / extended abstract track*)

- **Renard, M.**, Miller, C., Frei, M., Chua, Y. X., Tan, P. M. S., Kyaw, G. M. M., Zhang, W., Seah, J. X. T., Fan, X., Zhang, H., Parkinson, T., Lo, J. C.-Y., Lee, J. K. W., & Schiavon, S. (2026). *Bedroom nighttime temperature, humidity and sleep duration in Singapore homes*. **Oral presentation.**
- Zhang, W., Frei, M., Chua, Y. X., **Renard, M.**, Fan, X., Chong, A., Parkinson, T., Lee, J. K. W., Schiavon, S., & Miller, C. (2026). *Just-in-Time Adaptive Interventions (JITAI) to improve indoor air quality in sleep environments: A pilot study*. **Oral presentation.**

- Mani, R., Tan, S. C. C., **Renard, M.**, Frei, M., Loke, J. M. Q., Leow, C. H. W., Tan, P. M. S., Parkinson, T., Lo, J. C.-Y., Schiavon, S., & Lee, J. K. W. (2026). *Higher night-time temperatures are associated with shorter sleep among outdoor workers in non-air-conditioned, multi-occupant dormitories in the tropics*. **Oral presentation**.

#### **NRF CREATE Symposium 2026** — Singapore, June 2026

- Poster presentation on dynamic bedroom temperature control — a 30-night field trial of a “down-rise” cooling profile across 18 participants.

#### **Project HEATS Symposium** — MedScience Library, National University of Singapore, July 2026

- Presentation and poster on the heat exposure workstream; contributed to programme design and facilitation.

#### **Heat Early Warning System Symposium** — Brisbane, Australia, August 2025

- **Renard, M.** (2025). *From generalised warnings to personalised guidance: Synthesizing evidence to inform heat health advisory selection*.

#### **The Nutrition Society Congress 2024** — Belfast, Northern Ireland, 2–5 July 2024

- **Renard, M.**, Kelly, D. T., Ní Chéilleachair, N., Lavelle, F., & Ó Catháin, C. (2024). *Optimising dietary behaviour: Can sports nutrition research contribute to improvements in population health?* **Invited lecture — UK Postgraduate Competition Winner**.

#### **The Nutrition Society Irish Section 33rd Postgraduate Conference** — Technological University of the Shannon, Ireland, 22 February 2024

- **Renard, M.**, Lavelle, F., Kelly, D. T., Ní Chéilleachair, N., & Ó Catháin, C. (2024). *Determinants of dietary intake in team sport athletes: A mixed-methods approach*.

#### **NEX Grand Depart Symposium** — University of Copenhagen, Denmark, 27 June – 1 July 2022

- **Renard, M.**, Kelly, D. T., Ní Chéilleachair, N., Lavelle, F., & Ó Catháin, C. (2022). *Evaluation of cooking and food skills in Irish team sport athletes*.

#### **International Sport & Exercise Nutrition Conference** — Newcastle University, UK, 16–19 December 2019

- **Renard, M.**, Anton-Solanas, A., Kelly, D. T., & Ó Catháin, C. (2019). *Sports nutrition knowledge of Gaelic football players*. Published abstract: *International Journal of Sport Nutrition and Exercise Metabolism*, **30**(3). <https://doi.org/10.1123/ijnsnem.2020-0065>

#### **Forthcoming**

#### **International Conference on Environmental Ergonomics (ICEE) 2026** — Kuching, Malaysia, 17–21 August 2026

- **Renard, M.**, Frei, M., Chua, Y. X., Parkinson, T., Lo, J. C.-Y., Miller, C., Lee, J. K. W., & Schiavon, S. (2026). *Near-body–room air temperature difference and total sleep time in Singapore*. Abstract AB59. **Oral presentation, Session 1, Mon 17 August**.
- Tay, V. L. L., Morrison, S. A., **Renard, M.**, Tan, P. M. S., Zain, A., Lam, L. Y., Low, S., Eriksson, J. G., & Lee, J. K. W. (2026). *Physiological and behavioural responses of children to continuous exercise in mild-to-severe tropical heat stress*. **Invited presentation**.
- Kyaw, G. M. M., **Renard, M.**, Fan, X., & Lee, J. K. W. (2026). *A systematic review of adverse nighttime heat thresholds and associated effects on sleep, comfort and health*. Abstract AB4. **Oral presentation, Session 7, Tue 18 August**.
- Loke, J. M. Q., **Renard, M.**, Alhadad, S. B., Tan, S. C. C., & Lee, J. K. W. (2026). *The effects of cold and ice-based beverage ingestion on performance- and safety-related outcomes: A systematic review*. Abstract AB60. **Oral presentation, Session 8, Tue 18 August**.
- Loke, J. M. Q., **Renard, M.**, Alhadad, S. B., Tan, S. C. C., & Lee, J. K. W. (2026). *The effects of cold and ice-based beverage ingestion beyond physiological and perceptual outcomes: A systematic review*. Abstract AB65. **Oral presentation, Session 8, Tue 18 August**.
- Wee, J., Morrison, S. A., **Renard, M.**, Tan, P. M. S., Leow, C. H. W., Leow, M. K. S., Eriksson, J. G., & Lee, J. K. W. (2026). *Heat tips the balance: Exertional heat stress impairs postural stability in older adults*. Abstract AB6. **Oral presentation, Session 14, Thu 20 August**.
- Mani, R., Tan, S. C. C., **Renard, M.**, Frei, M., Loke, J. M. Q., Leow, C. H. W., Tan, P. M. S., Parkinson, T., Lo, J. C.-Y., Schiavon, S., & Lee, J. K. W. (2026). *Affordable and sustainable solutions to improve sleep among outdoor workers in non-air-conditioned, multi-occupancy dormitory in the tropics*. Abstract AB25. **Oral presentation, Session 16, Thu 20 August**.

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## MEDIA & PUBLIC ENGAGEMENT

Research from Project HEATS was featured across national media following the **NRF CREATE Symposium 2026**, with the work cited by the Chairman of the National Research Foundation Singapore.

- **The Straits Times** (25 June 2026) — “*What’s the temperature in the bedroom in S’pore? A muggy 28 deg C, says new study on residents*”. Interviewed and quoted on establishing context-specific sleep temperature guidance for tropical climates.
  - **Lianhe Zaobao** 联合早报 (25 June 2026) — “宿舍升温 1 摄氏度 客工平均少睡 10 分钟：研究团队探讨多项降温方案” — feature on dormitory temperature and sleep among outdoor workers.
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## TECHNICAL & METHODOLOGICAL SKILLS

**Statistical modelling** — Linear mixed-effects models (random intercepts and slopes); Mundlak within/between decomposition; fixed-effects OLS with cluster-robust standard errors; piecewise and segmented threshold models with AIC-based knot search; restricted cubic splines; generalised linear models (logistic, Poisson, negative binomial); GEE; survival analysis; cluster bootstrap; family-wise multiplicity control; hierarchical and multiple regression; thematic and mixed-methods analysis.

**Evidence synthesis** — PRISMA 2020 and PRISMA-P; PROSPERO registration (four records); systematic and scoping review methodology; MMAT; Cochrane Risk of Bias 2.0; meta-analysis appraisal including multilevel sensitivity models, publication bias assessment and certainty-of-evidence grading.

**Programming & tools** — Python (statsmodels, pandas, NumPy, seaborn, matplotlib, python-docx, PyMuPDF); R (ggplot2, tidyverse); SPSS; Git and GitHub; Bash; Quarto/HTML; reproducible reporting pipelines. Contributor to the internal [heatsviz](#) visualisation package; author of the open-access [CLIF statistical test chooser](#).

**Physiological measurement** — Wearable and in-home environmental monitoring (temperature, humidity, CO<sub>2</sub>, particulates); actigraphy and sleep outcome derivation (TST, WASO, sleep efficiency); core and skin temperature; thermal comfort and perceptual scales; VO<sub>2</sub>max and sub-maximal cardiopulmonary exercise testing; lactate threshold; resting metabolic rate and energy expenditure; anthropometry and body composition; functional movement screening; 12-lead ECG; spirometry; phlebotomy and point-of-care blood analysis.

**Languages** — English (native).